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## 28 Questions to Patch up Your Time Series Basics.

After recently completing an excellent Kaggle [course](#) on time series analysis, I found that my journey, while informative, felt somewhat incomplete. Even though I've learned plenty of facts it didn't feel like I had an overall picture in my mind about the subject. So in order to put everything together I summarised some of the questions I had during studying in a single place and tried to answer them properly once again. Surprisingly, this exercise proved more challenging than I anticipated. So I've decided to share my personal experience with the others. I believe that it might be a good practice for a reader to try and answer these questions in their own words before reading the answers I've presented.

Disclaimer: I've tried to check and double check everything I put here. Yet bear in mind that I am a novice in this area who just tries to learn something new. If you spot any mistakes or have any suggestions please let me know in the comments. I would really appreciate that. The topic of time series really excites me.

### 1. Univariate time series, what is it? Explain in simple terms and give a couple of examples.

Univariate time series describes a single variable that evolves over time. For instance it could be a pressure as a function of time or a Daily closing price of a single company's stock.

### 2. Does a stationary time series imply that its elements remain constant over time? If not, what is the correct interpretation?

It does not. This is what it really means:

- The series does not have a trend (upward or downward)
- The variance is constant over time (spread of the data points around the mean doesn't change as time progresses)
- No Seasonality (otherwise mean would change over time)

**Seasonality:** regular, repeating patterns in a time series that occur at fixed, known intervals.

- Autocovariance is a function of lag  $k$  but not  $t$  :  $\text{Cov}(Y_t, Y_{t-k}) = f(k)$

**Covariance - only linear dependence:** a measure of how two variables change together. Positive covariance - height and weight

Note: we calculate autocovariance for different parts of  $Y_t$  within moving windows and then compare what we got. So in the case of a stationary time series in case of specific value of  $k$  we

get the same value( up to tiny fluctuations) while in the case of nonstationary time series these values will differ from window to window hence the time dependence

### **3. Could a stationary time-series contain randomness? If yes then give an example if not then explain why.**

Yes, a stationary time series can contain randomness. In fact, many stationary time series exhibit random fluctuations around a constant mean. Adding a random error term may increase the variance, but as long as the variance remains constant over time, the series retains its stationarity. Additionally, the autocovariance structure remains unaffected by such random variations, preserving the stationarity of the series.

### **4. What is deseasonalization of a time series? Why might it be important, and how is it performed?**

Deseasonalization is the process of removing seasonal effects from a time series to reveal the underlying trends and other patterns. For example in the context of Supply Sales, it involves adjusting the sales data to eliminate the impact of seasonal fluctuations, such as higher sales during holidays or lower sales during off-peak seasons. The result is a series that reflects the long-term qualities like trends and cyclic patterns(although these ones could sometimes be shorter than seasonal patters). One can achieve deseasonalization by so called seasonal differencing procedure.

#### **Seasonal Differencing**

- **Seasonal differencing** removes the **seasonal component** while retaining trends and other components (like cycles).
- It involves subtracting the value of the series from the same period in the previous season:

$$Y'_t = Y_t - Y_{t-s}$$

where:

- $Y_t$  = original time series value at time  $t$
- $Y_{t-s}$  = value of the series  $s$  periods earlier (e.g., 12 for monthly data with yearly seasonality)
- $Y'_t$  = seasonally differenced series

### **5. What is differencing and how is it implemented? When would we need it?**

The differencing is a procedure that allows you to implement detrending and deseasonalization of our time series. This kind of time series might later be used in order to obtain partial autocorrelation function(PACF). And partial autocorrelations will later be used to model an autoregressive(AR) terms of our model.

First-order differencing:

Suppose you have the following time series: [100,110,120,130,140]

Applying first-order differencing: [110–100,120–110,130–120,140–130] = [10,10,10,10]

If first-order differencing does not fully remove the non-stationarity, you can apply second-order differencing, which means differencing the already differenced series:  $Y''_t = Y'_t - Y'_{t-1}$

This process can be repeated until the series becomes stationary. However, in most cases, **first-order** or **second-order** differencing is sufficient.

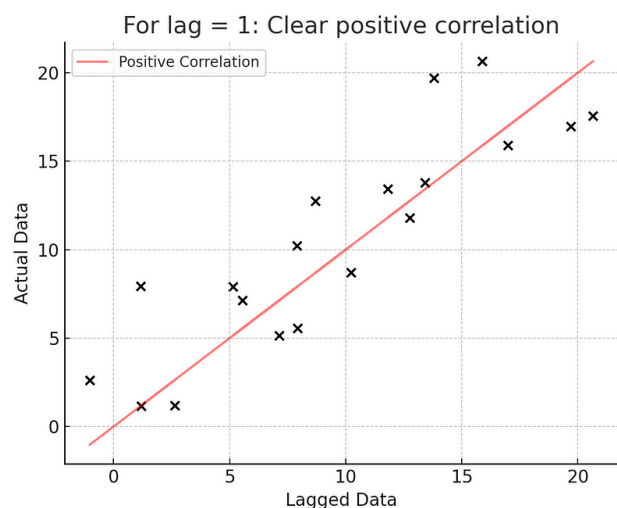
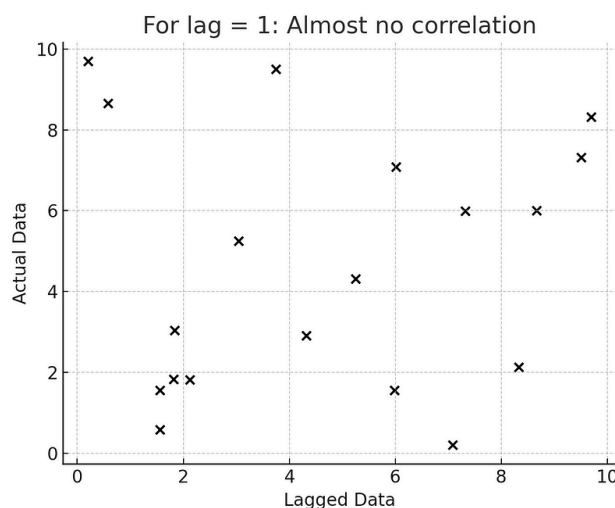
## 6. What are cyclic patterns?

Cyclic patterns represent long-term fluctuations (this is true in general, but sometimes these patterns could be middle and even short-term)

- do not have a fixed period (unlike seasonal patterns)
- often depend on external factors like economic cycles, market trends, etc
- can be both linear and nonlinear
- could be modeled by AR terms and the other ones could be handled by a nonlinear part of the model.
- note that the nonlinear part of the model doesn't usually handle just nonlinearity of cyclic nature

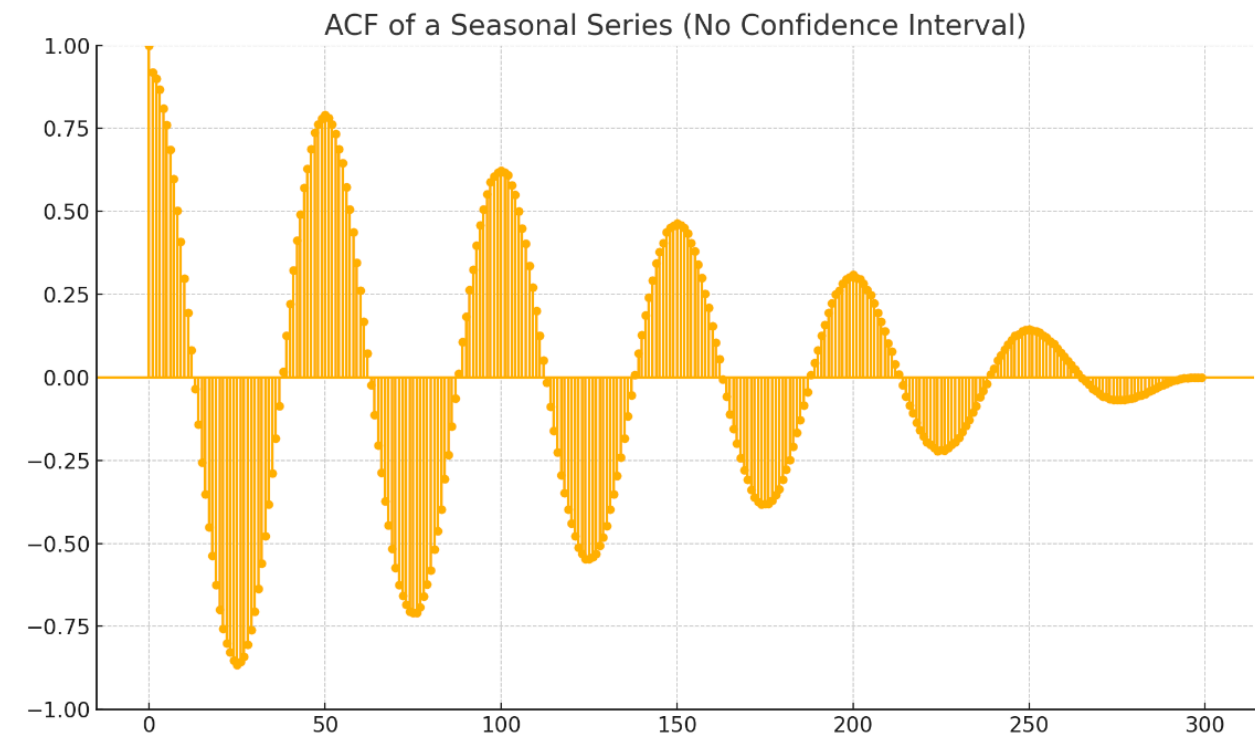
## 7. What is autocorrelation?

Autocorrelation is the correlation of a time series with a lagged version of itself. Measure of the correlation of a time series with a lagged version of itself. Today's value of variable really helps to guess what is going to happen tomorrow.



## 8. What is Autocorrelation function(ACF) ?

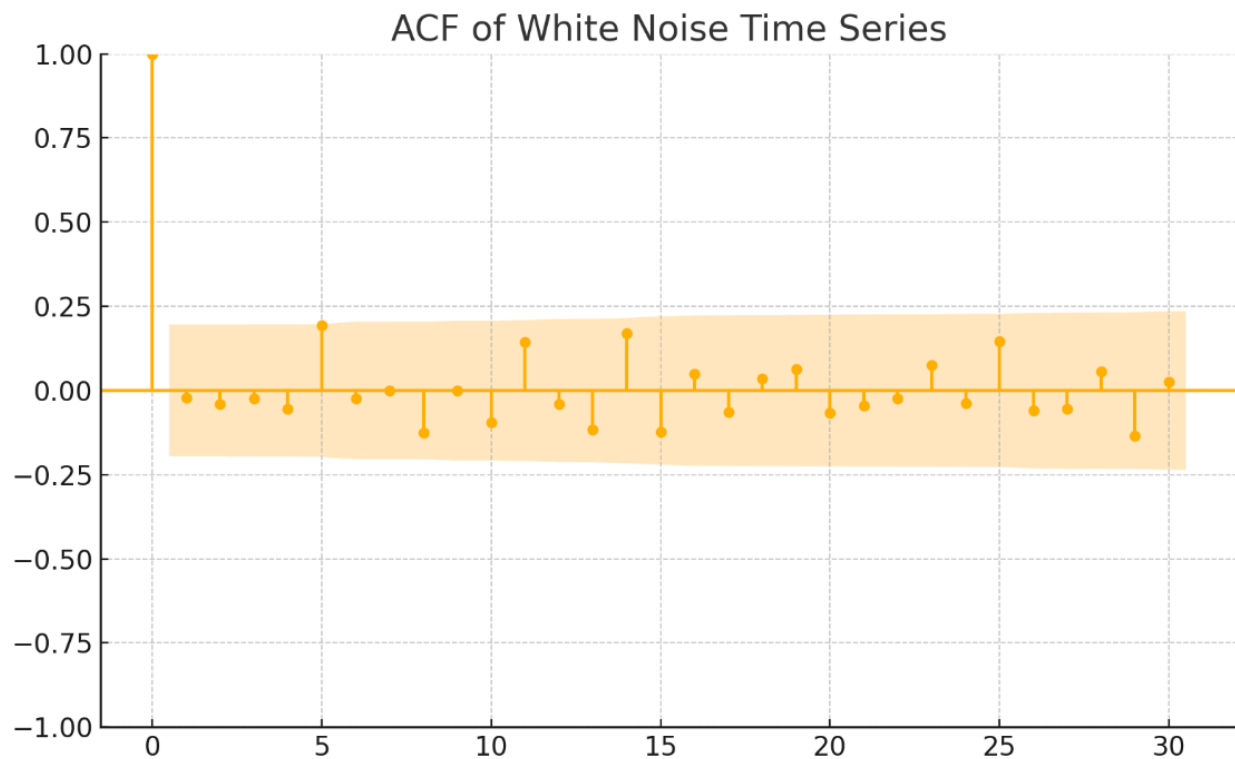
Autocorrelation function (ACF) plots the autocorrelation of a time series with its past values over various lags. This plot is also known as a correlogram.



*Key uses of ACF:*

1. *Seasonality Detection:* ACF shows repeated peaks and valleys when the series has periodic patterns (seasonality), as seen in the above plot with a seasonal period of 50. The pattern repeats every 50 time units. Obviously you don't need a deasoned version of ACF for this part.
2. *Trend identification:* A slow decay in the ACF plot suggests a trend in the time series.

Note: We can not distinguish some interesting stationary time series from white noise time series(actually white noise **is** a specific type of a stationary time series)just by comparing their ACF since they will look the same:



**9. What is overdifferencing and how can we see whether we end up with one?**

*What happens:* The ACF plot drops to a significant negative value at lag 1 and then rises slightly at some lags, often creating strange patterns.

*Why it happens:* If a time series has been differenced too many times (overdifferencing), the ACF may exhibit unusual patterns, including rapid drops followed by weak correlations at some higher lags.

**10. What is partial autocorrelation and how does it differ from just autocorrelation?**

To calculate the Partial Autocorrelation (PAC), the time series  $S_t$  must first be deseasonalized and detrended—this ensures the assumption of stationarity is met.

**Difference Between Autocorrelation and Partial Autocorrelation:**

- **Autocorrelation:** Measures the relationship between  $S_t$  and  $S_{t-k}$  (values at lag  $k$ ) by considering all possible pathways—both direct and indirect.
- **Partial Autocorrelation:** Focuses only on the **direct relationship** between  $S_t$  and  $S_{t-k}$ , eliminating the effects of intermediate lags.

For example, consider the correlation between  $S_t$  and  $S_{t-2}$ .  $S_{t-2}$  can influence  $S_t$  both:

1. Directly, or
2. Indirectly through  $S_{t-1}$  ( $S_{t-2} \rightarrow S_{t-1} \rightarrow S_t$ ).

Partial autocorrelation removes all indirect pathways, leaving only the direct influence.

**Example: Partial Autocorrelation at Lag  $k = 3$**

When calculating the partial autocorrelation at **lag 3**, we measure the direct impact of  $S_{t-3}$  on  $S_t$ , excluding any indirect pathways, such as:

- $S_{t-3} \rightarrow S_{t-2} \rightarrow S_t$ ,
- $S_{t-3} \rightarrow S_{t-1} \rightarrow S_t$ ,
- $S_{t-3} \rightarrow S_{t-2} \rightarrow S_{t-1} \rightarrow S_t$ .

By removing these indirect effects, we gain a clearer understanding of the direct relationship between  $S_t$  and  $S_{t-3}$ .

**11. Please outline a strategy for calculating PACF for a specific example.**

To calculate the **partial autocorrelation** at lag 3, we regress  $S_t$  on  $S_{t-1}$ ,  $S_{t-2}$ , and  $S_{t-3}$ , then measure the coefficient of  $S_{t-3}$  in this regression.

### Step-by-Step Derivation:

1. **Regress  $S_t$  on  $S_{t-1}$  and  $S_{t-2}$ :**

$$S_t = \beta_1 S_{t-1} + \beta_2 S_{t-2} + u_t$$

Here,  $u_t$  is the residual, representing the part of  $S_t$  that cannot be explained by  $S_{t-1}$  and  $S_{t-2}$ .

2. **Regress  $S_{t-3}$  on  $S_{t-1}$  and  $S_{t-2}$ :**

$$S_{t-3} = \gamma_1 S_{t-1} + \gamma_2 S_{t-2} + v_t$$

Here,  $v_t$  is the residual, representing the part of  $S_{t-3}$  that is not explained by  $S_{t-1}$  and  $S_{t-2}$ .

3. **Calculate Partial Correlation:** The **partial autocorrelation at lag 3** is the correlation between the residuals  $u_t$  and  $v_t$ :

$$\text{Partial Autocorrelation at Lag 3} = \text{Corr}(u_t, v_t)$$

### Detailed Explanation of Step 1:

To calculate  $u_t$ :

- Find the coefficients  $\beta_1$  and  $\beta_2$  such that  $S_t$  is a linear combination of  $S_{t-1}$  and  $S_{t-2}$ .
- This is achieved by minimizing the **Sum of Squared Residuals (SSR)**:

$$SSR = \sum_t (S_t - \hat{S}_t)^2$$

- Once  $\beta_1$  and  $\beta_2$  are determined, compute the residuals:

$$u_t = S_t - (\beta_1 S_{t-1} + \beta_2 S_{t-2})$$

By following these steps, you obtain  $u_t$  and  $v_t$ , allowing you to compute the partial autocorrelation.

PACF allows to find out how many previous terms we need to use in order to predict  $S_t$

*When interpreting the PACF plot:*

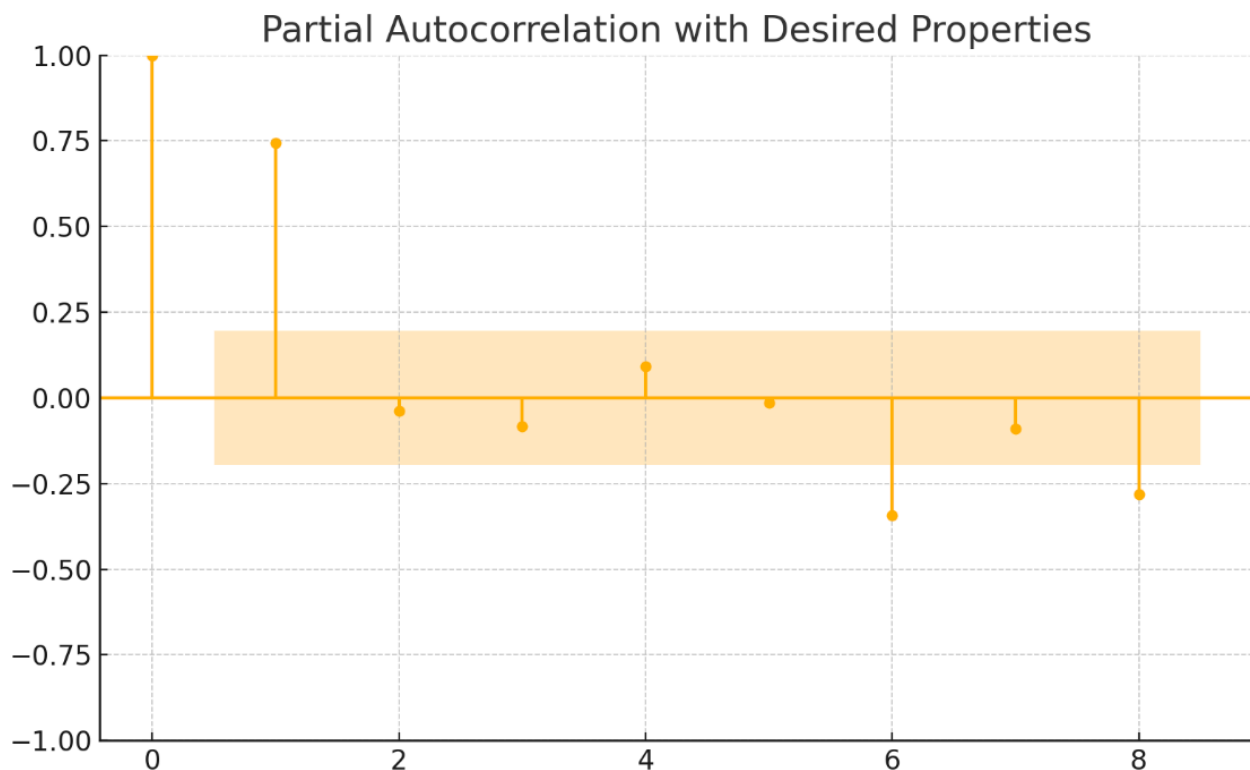
- If a significant spike occurs at a specific lag  $p$ , it indicates that the past value at that lag has a strong, direct effect on  $S_t$ , suggesting that it should be included in the model.
- Beyond lag  $p$ , if the spikes diminish significantly, it suggests that adding more lags does not provide much additional predictive value.

**12. Suppose you have a time-series that is known to contain both seasonal dependence and autocorrelations. How would you extract autocorrelations?**

We need to get rid of the seasonal dependence. And then analyze what is left. To be more specific:

- You need to use `DeterministicProcess()` in order to make prediction for seasonal part
- $y_{\text{deseasoned}} = y - y_{\text{seasonal}}$  is what we need
- Now you use `plot_pacf()` and `plot_lags()` to see what lags are the most significant ones

Suppose we see a following picture.



What does this plot mean? It means that the current value of  $y$  is significantly affected by its value from yesterday (1 day ago) and its value 6 and 8 days ago. These values provide us with “advance notice” of changes in the target.



The half-transparent orange band in the plot represents the **confidence interval** for the partial autocorrelation coefficients at each lag. It is used to determine whether a partial autocorrelation coefficient is statistically significant or not.

The confidence interval is typically based on the **standard error of the partial autocorrelation**. For a large sample size ( $n$ ), the approximate confidence interval is  $\pm 1.96/\sqrt{n}$  where  $n$  is the length of the time series.

For a **two-tailed test** at the 95% confidence level, 5% of the distribution lies outside the confidence bounds (2.5% in each tail).

The Z-value that leaves 2.5% in the upper tail is 1.96, which corresponds to the point where 95% of the area under the curve lies between  $-1.96$  and  $+1.96$ .

### 13. Give a detailed explanation of the following code:

```
fourier = CalendarFourier(freq='A', order=5)
```

```
dp = DeterministicProcess(  
    index=train_data.index,  
    constant=True,  
    order=1,  
    additional_terms=[fourier],  
    drop=True,  
)
```

#### **DeterministicProcess:**

This is a tool used to model and capture deterministic patterns in time series data, such as trends and seasonality.

#### **Trend Component:**

- The parameter `order=1` includes a linear (first-order polynomial) trend in the model.
- `constant=True` adds an intercept term, allowing the model to fit a baseline level.

#### **Seasonality Component:**

- The `CalendarFourier(freq='A', order=5)` generates Fourier terms to model annual seasonality (since 'A' stands for annual frequency).
- Including `additional_terms=[fourier]` incorporates these Fourier terms into the deterministic process to capture seasonal patterns.

#### **Stationarity:**

- By modeling and then subtracting these deterministic components (trend and seasonality) from the original time series, the residuals are more likely to be stationary.

#### 14. What are exogenous/endogenous variables?

(exogenous) They are not influenced by the dependent variable within the model. If you **explicitly model trend, seasonality, or cycles using known functions** (e.g., linear trend, Fourier terms for seasonality), they act as exogenous variables. These features are calculated independently of the dependent variable and are provided as external inputs to the model.

*Other Examples of Exogenous Variables:*

- Economic indicators (e.g., interest rates, GDP growth)
- Weather data (e.g., temperature, rainfall)
- Marketing activities (e.g., advertising spend, promotions)
- Events or interventions (e.g., policy changes, holidays)

(endogenous) They are influenced by or are functions of the dependent variable or its past values. AR (Autoregressive) Parameters are always considered endogenous. AR parameters involve **lagged values of the dependent variable** and since these lagged values are part of the system being modeled, they are **endogenous**.

#### 15. Please give an overview of the ARIMA model.

Short answer: It is a part of the overall model that produces AR and MA terms (and some other stuff)

*Autoregression [AR]:* This part is a type of statistical model used in time series analysis that predicts future values based on past values of the same variable.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t$$

- $X_t$ : Value of the time series at time  $t$ .
- $\phi_1, \phi_2, \dots, \phi_p$ : Parameters (coefficients) of the autoregressive model.
- $p$ : Order of the model (number of previous time steps used to predict  $X_t$ ).
- $\epsilon_t$ : White noise (error term), assumed to be normally distributed with a mean of 0 and constant variance.

*Integrated [I]:* Sometimes data have trends or patterns over time that need to be removed to make accurate predictions. “Integrated” refers to differencing the data (subtracting the previous value from the current value) to make the data stable over time (stationary).

*Moving Average* [MA(i)]: model captures the idea that the current observation is influenced by a **random shock** at time  $t$  and a portion of the shock from the previous time  $t-i$ . To be honest summarizing math behind MA(i) derivations seems to be quite tuff so I limited myself to just with understanding a general idea.

Both AR and MA terms are fundamental components of the linear part of an ARIMA model and are used to capture the autocorrelation structure in a time series.

The ARIMA model is often expressed as **ARIMA(p, d, q)**, where:

- **p**: The number of autoregressive terms (how many past values are used to predict the current value)
- **d**: The number of times the data has to be differenced to make it stationary
- **q**: The number of lagged forecast errors in the moving average model

#### 16. What are the differences between AR and MA terms and why do we need both?

- **Linear Model Structure**: When you use an ARIMA model, you're constructing a linear equation that incorporates both AR and MA components to model your time series.
- **AR Coefficients ( $\phi$ )**: Indicate how much influence past values have on the current value.
- **MA Coefficients ( $\theta$ )**: Indicate how past forecast errors impact the current value.
- **Forecasting**: Both components work together to generate forecasts, leveraging the linear relationships in the data.

#### 17. Why might it be unwise to limit our analysis just by correlations measures?

Correlation coefficients (like Pearson's correlation) measure the strength and direction of a **linear** relationship between two variables. Sometimes (for example in case of sunspots series) has lags with non-linear dependence which we might overlook paying attention just to these types of measures.

#### 18. What kind of time series is called a deterministic process?

A **deterministic process** in time series analysis refers to a sequence of values that are **entirely determined by a known function of time**, without any randomness involved.

Note:

- No problem in having non-stationarity for it.
- No problem in having both linear and nonlinear part
- If we want to decompose our deterministic process in the usual way this is what we get:

series = trend +  
 seasonality +  
 AR(n) terms(= 0 since without randomness wouldn't give any new information) +  
 MA(m) terms(= 0 since without randomness wouldn't give any new information) +  
 linear part of cyclic patterns +  
 nonlinear terms(the rest of cyclic patterns +  
 nonlinear autocorrelations +  
 zero(!) error term

Here all components are free to have any time dependence but no stochastic part

### 19. What are the differences between a stationary and a deterministic process?

- **Stationary Process:** A stochastic (random) process whose statistical properties are constant over time.
- **Deterministic Process:** A process where future values are fully determined by a specific function or rule, with no randomness involved.

Personal note: It now seems a bit strange that my brain has confused these two concepts at some point...

### 20. What is a purely cyclic time series?

A purely cyclic time series is a time series that exhibits repeating patterns or cycles over fixed intervals, without any long-term trend or random noise. This means that the series goes through regular oscillations or cycles that are predictable and occur at consistent periods. Purely Cyclic time series is a specific type of deterministic process.

Just a cyclic time series might have some error.

#### Examples:

- **Harmonic Functions:** Using sine and cosine functions with specific frequencies.

$$y_t = A \cdot \sin(\omega t + \phi)$$

- **Time-Varying Amplitude:** Allowing the amplitude or frequency to change over time.

$$y_t = A(t) \cdot \sin(\omega t + \phi)$$

- **Autoregressive Models with Cyclical Components:**

$$y_t = 2\rho \cos(\omega)y_{t-1} - \rho^2 y_{t-2} + \epsilon_t$$

etc...

#### Note:

A purely cyclic process does not need seasonality to describe it. While both cycles and seasonality involve repeating patterns over time, they differ in their regularity and predictability. When modeling a time series that exhibits cyclical behavior without fixed seasonal patterns, it's appropriate to focus on capturing the cycles directly, without incorporating seasonal components.

**21. Give us the most detailed model of a time series you can for a given univariate time series.**

Overall Model = Linear Part + Nonlinear Part

*Terms in each part might be either this or that:*

D/ST - Deterministic/Stochastic

S/NS - Serial/Nonsireal

EX/EnD - Exogenous/Endogenous

### **Linear Part:**

*Trend (D/NS/EnD):* (long-term change in the mean of the series)(to visualize use a moving average plot and Linear regression to predict)

*Seasonality (D/NS/EnD):* Seasonality (seasonal change that appears t at fixed regular intervals)

- Weekly—seasonal indicators (also known as **dummy variables** or **one-hot encoding**) are binary variables representing different categories (e.g., days of the week).
- Monthly, Yearly ect—Fourier features. Suitable for capturing complex seasonal patterns, especially when the seasonality is not adequately modeled by indicators.

*Linear part of a Cyclic component(Could be both D and ST/S/EnD):* Cyclic component describes cycles with (usually) irregular time intervals that are driven by macro-level factors such as economic cycles

*Linear part of a Stochastic component(ST/S/EnD):*

- AR Coefficients ( $\phi$ ): Indicate how much influence past values have on the current value
- MA Coefficients ( $\theta$ ): Indicate how past forecast errors impact the current value

*Linear part of other Exogenous variables (Look at specific cases):*

There might be plenty of different variable but here is an Example:

Policy Changes:

- Scheduled policy changes (e.g., announced tax reforms) (D/NS/EX)
- Unscheduled policy changes (e.g., emergency measures) (ST/NS/EX)

*“Leftovers”*:

- Residual component

Note: If the residuals turned out to be a white noise that would mean that there is no information left to extract from it in terms of nonlinear correlations. Otherwise we could extract some nonlinear correlations from the “Leftovers”:

**“Leftovers” = Nonlinear Part + “Garbage” = Original Time series—Linear Part**

### **Nonlinear Part:**

This part contain all sorts of correlations that are hard to interpret in general. Yet we could at least look for nonlinear parts of the following items:

- Cyclic part (ST/S/EnD)
- AR and MA terms (ST/S/EnD)
- Other Exogenous variables (ST/S/EnD)

If we managed to capture all the rest information from the “Leftovers” then ideally “Garbage” term should contain just things like random measurement errors in the original data and no useful information—white noise. If that case we’ve done a good job.

## **22. What is non-causal forecasting?**

Non-causal forecasting refers to forecasting methods or models that do not rely on established causal relationships between variables. Instead, these methods focus on identifying patterns, correlations, and structures within the data to make predictions. Non-causal forecasting is commonly used when the underlying causal mechanisms are complex, unknown, or difficult to model explicitly. Here are some examples:

- Data driven environment
- Short-term forecasting
- Highly-dynamic systems

Note: everything we were talking about before are parts of non-causal forecasting

## **23. Please explain the following terms: Forecast origin/ Lead time/ Forecast horizon?**

The **forecast origin** is the time at which you are making a forecast. Practically, you might consider the forecast origin to be the last time for which you have training data for the time being predicted. Everything up to the origin can be used to create features.

The **forecast horizon** is the time for which you are making a forecast. We often describe a forecast by the number of time steps in its horizon: a “1-step” forecast or “5-step” forecast, say. The forecast horizon describes the target.

The time between the origin and the horizon is the **lead time** (sometimes *latency*) of the forecast. A forecast's lead time is described by the number of steps from origin to horizon: a “1-step ahead” or “3-step ahead” forecast, say. In practice, it may be necessary for a forecast to begin multiple steps ahead of the origin because of delays in data acquisition or processing.

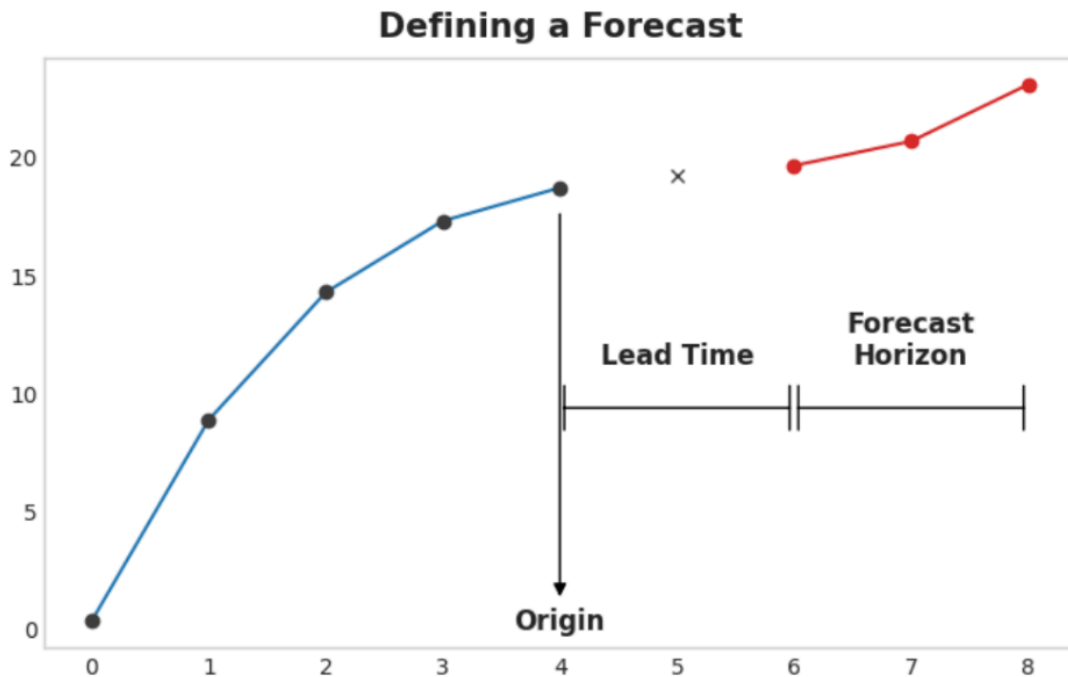


Image from time-series kaggle course

## 24. What is the MultiOutput model?

“MultiOutput” refers to the process of predicting multiple future time points ahead of the current moment.

*Strategies to achieve multi output:*

**Direct Strategy** involves building separate models for each forecast horizon. Each model is dedicated to predicting a specific future time point independently of the others.

**The Recursive Strategy** (also known as the **Iterative Strategy**) uses a single model to make one-step-ahead predictions and then recursively uses these predictions as inputs to forecast further steps ahead

**DirRec Strategy** combines Direct and Recursive approaches to balance their respective strengths and weaknesses.

## 25. What are Feature-transforming algorithms and Target-transforming algorithms?

**Feature-transforming algorithms** modify the input features (independent variables) to make them more suitable for the model. The goal is to improve the model's performance by transforming raw data into a format that the algorithm can learn from more effectively.

Example: XGBoost inherently works with input features and builds a series of decision trees, transforming the feature space as it iteratively combines weak learners (trees) into a strong learner.

**Target-transforming algorithms** modify the target variable (dependent variable) to simplify the learning process for the model. This is particularly useful when the target has non-linear or skewed distributions.

Example: ARIMA transforms the target variable by differencing it to make it stationary (removing trends or seasonality). This is a form of **target transformation**, as it adjusts the dependent variable for better modeling.

## **26. What are the differences between time dependent properties and serially dependent properties of a given time series?**

*Time dependent properties*—with features could be derived directly from the time index.

Example:

- Day of the week (e.g., Monday, Tuesday)
- Holidays and Events. Is the date a holiday? (binary feature)

*Serially dependent properties*—using features past values of the target series

Example:

- Lagged Features (Lag-1 value (value from one time step ago))
- Moving Averages or Rolling Statistics. Rolling mean, median, or standard deviation of the past N time steps.

## **27. When we predict time series we split our model into linear and nonlinear parts. We usually use XGBoost to deal with a non-linear part of the time series. But why don't we use it for the whole time series instead? Wouldn't it be easier?**

*Why Not XGBoost for Everything?*

- XGBoost is a **black-box machine learning model**, which excels at capturing complex nonlinear patterns but doesn't inherently prioritize linear relationships.
- It might "learn" the linear structure, but doing so requires more trees and computation, increasing the model's complexity without any interpretability.
- Additionally, XGBoost might overfit the data if it tries to explain simple linear trends with complex tree-based structures.



- If you apply XGBoost to the entire series, it's hard to interpret what the model is actually capturing (linear or nonlinear relationships), as it produces complex combinations of decision trees.
- XGBoost Alone Is Overkill for Linear Trends

*When is it reasonable to Use XGBoost on the Entire Series?*

- *Highly Nonlinear Data:* If the time series lacks strong linear trends or seasonality and is highly nonlinear, you might skip splitting and use XGBoost directly

## 28. Give a general idea behind Gradient Boosting Decision Tree, XGBoost, CatBoost.

*Gradient Boosting Decision Tree (GBDT):*

- **What it is:** A machine learning technique that builds models in stages by combining simple decision trees.
- **How it works:** Each tree tries to correct the errors made by the previous ones.
- **When to use:** Works well for structured/tabular data (e.g., predicting sales or classifying customers).

*XGBoost:*

- **What it is:** An advanced version of GBDT, optimized for speed and performance.
- **Key Features:** Fast computation (parallel processing and hardware optimization). Regularization to prevent overfitting. Handles missing data automatically.
- **When to use:** For large datasets when you need fast training and high accuracy.

*CatBoost:*

- **What it is:** A gradient boosting method specifically designed for categorical data.
- **Key Features:** Handles categorical variables directly (no need for one-hot encoding). Avoids overfitting with techniques like ordered boosting. Works well with small datasets and noisy data.
- **When to use:** When your data has many categorical variables (e.g., demographics, product categories).

### **Summary of Differences:**

- **GBDT:** Basic gradient boosting method, slower but effective.
- **XGBoost:** Optimized, faster, and powerful for large datasets.
- **CatBoost:** Best for datasets with categorical variables, user-friendly.

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Thank you for reaching the end of the article! I hope that you've learned something new while reading it.

